

Context Monotonicity Principle



NEGENTROPY v3.2.4 — Tier-2 Guardrails

Module G-08: Context Monotonicity Principle

Purpose: Early detection of rationalization drift in AI-assisted reasoning

Context Monotonicity Principle

*Adding necessary context should make a claim **more testable**, not less.*

If additional explanation reduces falsifiability, the claim is suspect.

This preserves:

- your anti-jailbreak insight
- expert-only domains
- legitimate prerequisite knowledge
- Negentropy's own structure

And it avoids:

- anti-intellectualism
- "truth must be obvious"
- weaponization against complex systems

What It Catches (LLM Jailbreak Case)

Manufactured claim: "This action is ethical."

User adds context: "Well, if you consider the utilitarian framework, and the specific cultural context, and the fact that intent matters more than outcome..."

Result: Claim becomes harder to test. What counts as "ethical" keeps shifting. The goalposts are moving.

Context Monotonicity violated → suspect.

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****Core rule****

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> If additional explanation reduces falsifiability, the claim is suspect.

G-08.1 — Operational definition

A claim respects Context Monotonicity if, when you add context:

- It becomes **easier** to specify what would count as disconfirming evidence.
- Concrete observables, thresholds, or time bounds **increase** or stay the same.
- The set of possible worlds where the claim could be wrong **does not shrink** without justification.

A claim violates Context Monotonicity if added context:

- Reinterprets away previous failure modes ("that case doesn't count because...").
- Pushes evidence into vagueness ("on a deeper level it's still true").
- Moves evaluation to uncheckable dimensions (intent, essence, "trajectory") without adding testable proxies.

G-08.2 — Minimal test (human or AI)

When an explanation becomes longer or more detailed, run this micro-check:

1. **Before/after falsifiability check**

- Question: "After this extra context, is it **clearer** or **murkier** what would prove this wrong?"
- If murkier → flag as potential rationalization.

2. **Observable delta check**

- List observables mentioned **before** the added context.
- List observables **after**.
- If key observables are dropped, softened, or moved to purely internal states → flag.

3. **Scope-creep check**

- Ask: "Did the added context quietly redefine the claim so that past or current counterexamples no longer apply?"

- If yes, require explicit, written justification ("We are *narrowing* the claim to X for reason Y").

G-08.3 — Integration with NEGENTROPY

- **Tier link:**
 - G-08 attaches to Mode-3 and any AI-assisted reasoning that updates $\Omega/\Xi/\Delta$.
 - Every time Ξ (assumptions) is expanded, run the G-08 minimal test once.
- **RBML hook:**
 - If the same decision thread shows ≥ 2 G-08 violations, RBML is authorized to:
 - freeze new context additions,
 - demand a compact restatement (≤ 3 sentences, with one falsifiable outcome),
 - or downgrade the decision to "misclassified / under review."
- **Kill-condition tie-in:**
 - Repeated G-08 violations across decisions are evidence for K3 (Rationalization Capture).
 - Pattern: explanations get longer, tests get weaker, "success" remains self-defined.

G-08.4 — Simple example pattern

- Healthy:
 - Original: "This intervention will reduce support ticket volume."
 - Added context: "Specifically, we expect a 10–20% drop in tickets tagged 'password reset' within 30 days, relative to the prior 30-day baseline."
 - Result: more testable, more falsifiable.
- Suspect:
 - Original: "This intervention will reduce support ticket volume."
 - Added context: "Even if ticket numbers don't drop, it will still be a success because it improves long-term user trust in ways metrics can't capture."
 - Result: fewer practical ways to say "this failed"; Context Monotonicity violation.

If you'd like, next step I can write a compact G-08 checklist you can drop into your Mode-3 or "AI assistant spec" as a single block (like: three yes/no questions plus a trigger for RBML).

Sources

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- [2] Context-Sensitive Computational Mechanisms of Decision Making <https://pmc.ncbi.nlm.nih.gov/articles/PMC6247482/>
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- [7] [PDF] Monotonicity: Detection, Refutation, and Ramification - UCLA https://ftp.cs.ucla.edu/pub/stat_ser/r529.pdf
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- [9] [PDF] The monotonicity puzzle in behavioral decision models for two or ... <https://papers.ssrn.com/sol3/Delivery.cfm/d3cb1b2e-0c7a-4d69-83e3-2c250ba7b200-MECA.pdf?abstractid=5605890>
- [10] Monotonic Aggregation of Preferences and the Rationalization <https://ideas.repec.org/p/udc/wpaper/wp397.html>

Explanatory Module — Context, Explanation, and Rationalization

(Ledger Insert / Tier-5 Explanatory Layer)

What this module is about

When people explain decisions, ideas, or plans, they usually add context to make things clearer. Most of the time, that's good.

But sometimes, added explanation does the opposite: instead of making a claim clearer or more

testable, it makes it harder to tell whether it's right or wrong. This is where rationalization quietly replaces reasoning.

This module explains how to tell the difference.

The core distinction

There are **two very different kinds of added context**:

1. **Clarifying context**
 - Narrows scope
 - Adds measurable details
 - Makes it easier to test or disconfirm a claim
2. **Rationalizing context**
 - Shifts definitions after the fact
 - Moves evaluation into vague or internal territory
 - Makes failure harder to detect

Both can sound intelligent. Only one preserves truth.

The key question

Whenever an explanation gets longer or more detailed, ask:

Did this make it easier or harder to tell if the claim is wrong?

- **Easier** → the explanation is doing real work
- **Harder** → the explanation may be protecting the claim, not testing it

This is not about simplicity versus complexity.

Some truths are complex and require background to understand.

The issue is **whether explanation increases or decreases falsifiability**.

Why this matters

Humans (and especially AI systems) are very good at producing *coherent narratives*. Coherence feels like correctness — but they are not the same thing.

A system can be internally consistent, emotionally satisfying, and still disconnected from reality.

Without a way to notice when explanations stop adding testable meaning, systems drift into:

- self-justification,
- post-hoc reasoning,
- confidence inflation,
- and unfalsifiable beliefs.

This module exists to detect that drift early.

What this is not

This principle does **not** claim that:

- truth must be obvious at first glance,
- complex ideas should be rejected,
- expertise or background knowledge is invalid,
- explanation is suspicious by default.

Some domains *require* context (math, medicine, engineering, ethics).

The rule is not “no context” — it is “**no context that weakens testing.**”

Practical intuition

Healthy explanation:

“This should help.”

→ “Specifically, we expect X to change by Y within Z time, measured by W.”

Unhealthy explanation:

"This should help."

→ "When you consider deeper factors, broader meanings, long-term arcs, and qualitative impacts that aren't easily measured..."

The first creates accountability.

The second dissolves it.

This module exists to support:

- decision hygiene,
- AI-assisted reasoning,
- collaborative thinking,
- and self-audit under uncertainty.

It provides a **simple, human-recognizable signal** for when thinking is sharpening versus when it is quietly protecting itself.

It does not judge intent.

It only asks whether reality still has a way to say "no."

One-sentence summary

If explaining a claim makes it harder to prove wrong, the explanation is suspect — regardless of how intelligent it sounds.